

15-mavzu. Transport masalasini yechish usullari. Transport masalasini yechishning potentsiallar usuli

Reja

1. Transport masalasi matematik modelini qurish.
2. Transport masalasini potentsiallar usulida yechish hamda bu yechimni amaliy dasturlar yordamida olingan yechimlar bilan solishtirish.
3. Eng yaxshi yechimni topish va bu yechim maqsad funksiyasi va cheklanish shartlarini qanoatlantirishini tekshirish.

Kirish

Transport masalasi chiziqli dasturlashning eng muhim va amaliy masalalaridan biri hisoblanadi. Ushbu masala iqtisodiyot, logistika, sanoat ishlab chiqarishi, energetika, qurilish va boshqa ko'plab texnik sohalarda keng qo'llaniladi. Transport masalasining asosiy mohiyati shundan iboratki, mavjud manbalardan (omborlar, zavodlar, ishlab chiqarish punktlari) iste'molchilarga (do'konlar, sexlar, obyektlar) mahsulotni eng kam xarajat bilan yetkazib berish talab etiladi.

Real hayotda transport xarajatlari katta ahamiyatga ega bo'lib, ularni minimallashtirish korxona samaradorligini sezilarli darajada oshiradi. Shu sababli transport masalasini matematik model ko'rinishida ifodalash, uni turli usullar yordamida yechish va eng yaxshi (optimal) yechimni aniqlash muhim ahamiyat kasb etadi.

Mazkur mavzuda transport masalasining matematik modeli, uni potentsiallar usuli yordamida yechish hamda olingan yechimning optimal ekanligini tekshirish masalalari batafsil va tushunarli tarzda yoritiladi.

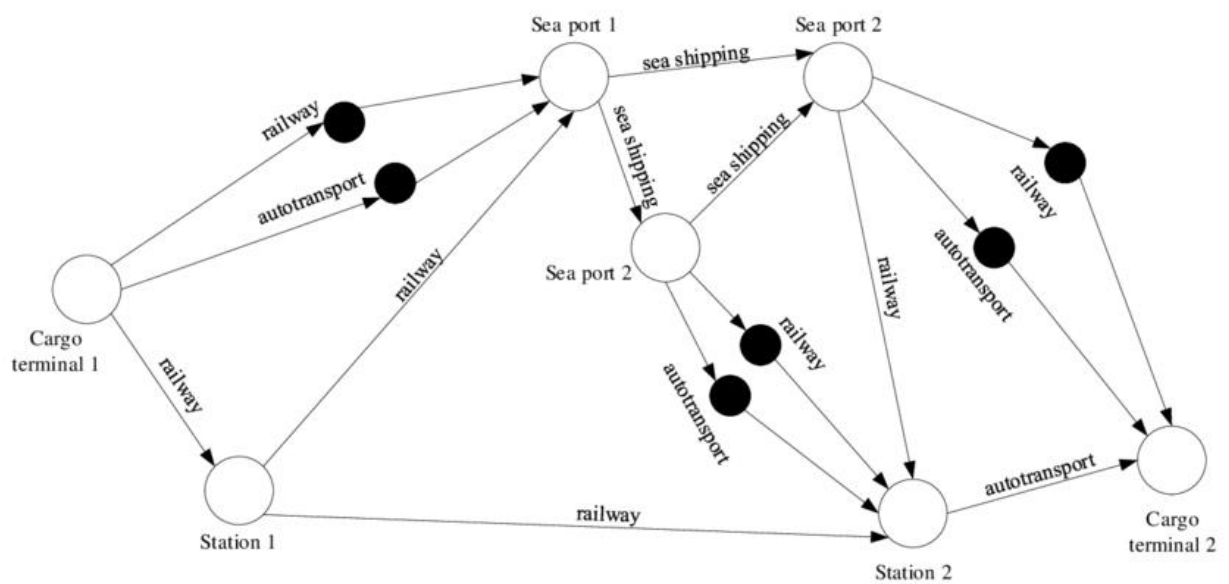
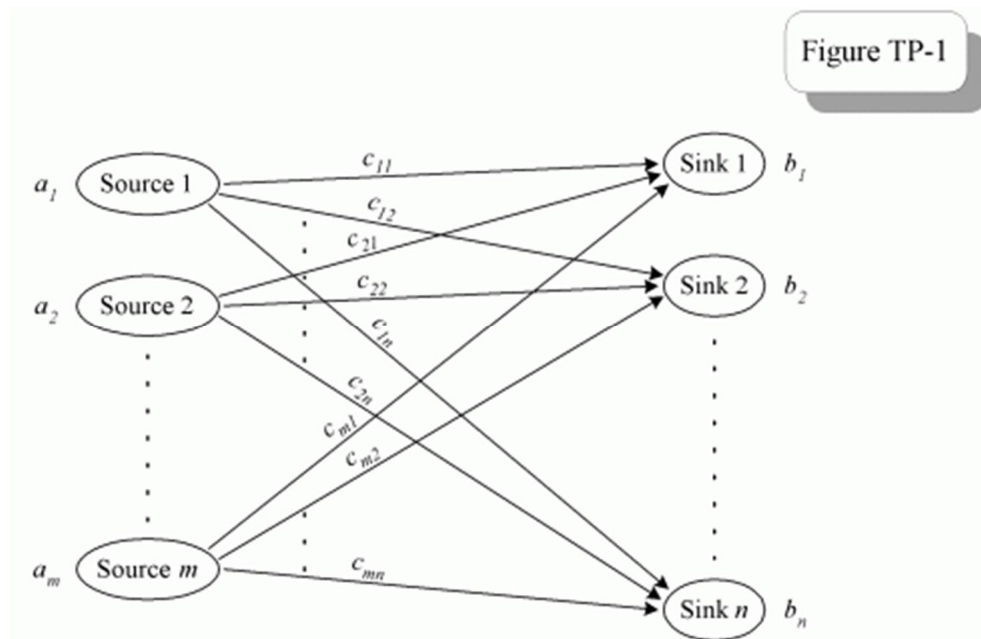
1. Transport masalasi matematik modelini qurish

1.1. Transport masalasining mohiyati

Transport masalasi quyidagi asosiy elementlardan tashkil topadi:

- **Manbalar (yetkazib beruvchilar)** – mahsulot jo'natiladigan punktlar;
- **Iste'molchilar (qabul qiluvchilar)** – mahsulot qabul qilinadigan punktlar;
- **Zaxiralar** – har bir manbada mavjud mahsulot miqdori;
- **Talablar** – har bir iste'molchiga kerakli mahsulot miqdori;
- **Transport xarajatlari** – mahsulotni bir birlik miqdorda tashish narxi.

Masalan, 3 ta zavod va 4 ta iste'molchi mavjud bo'lsin. Har bir zavoddan har bir iste'molchiga mahsulot tashish xarajati oldindan ma'lum.



1.2. Matematik model elementlari

Faraz qilaylik:

- a_i — i -manbadan mavjud mahsulot miqdori
- b_j — j -iste'molchining talabi
- c_{ij} — i -manbadan j -iste'molchiga tashish xarajati
- x_{ij} — i -manbadan j -iste'molchiga jo'natiladigan mahsulot miqdori

1.3. Maqsad funksiyasi

Transport masalasining maqsadi — **umumiy transport xarajatlarini** Transport masalasining maqsadi — **umumiy transport xarajatlarini minimallashtirish:**

$$Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \rightarrow \min$$

Bu funksiya barcha manbalar va iste'molchilar bo'yicha tashish xarajatlarining yig'indisini ifodalaydi.

1.4. Cheklanish shartlari

1. Manbalar bo'yicha cheklanish

Har bir manbadan jo'natilgan mahsulot miqdori uning zaxirasidan oshmasligi kerak:

$$\sum_{j=1}^n x_{ij} = a_i$$

1. Iste'molchilar bo'yicha cheklanish

Har bir iste'molchining talabi to'liq qondirilishi kerak:

$$\sum_{i=1}^m x_{ij} = b_j$$

1. Manfiy bo'lmaslik sharti

$$x_{ij} \geq 0$$

Agar umumiy zaxira umumiy talabga teng bo'lsa, masala muvozanatlashgan transport masalasi deyiladi.

2. Transport masalasini potentsiallar usulida yechish va dasturiy yechimlar bilan solishtirish

2.1. Potentsiallar usuli tushunchasi

Potentsiallar usuli — bu transport masalasining optimal yechimini tekshirish va yaxshilash uchun qo'llaniladigan klassik usuldir. Ushbu usul boshlang'ich reja asosida ishlaydi va xarajatlarni yanada kamaytirish imkoniyati mavjud yoki yo'qligini aniqlaydi.

Potentsiallar usuli ko'pincha:

- Shimoliy-g'arbiy burchak usuli,
 - Minimal xarajatlar usuli
- kabi usullar yordamida topilgan boshlang'ich yechimni optimallashtirishda ishlatiladi.

		Destination					
		D1	D2	D3	D4	Supply(S _i)	
Source	O1	200	50			250	
		3	1	7	4	50	0
	O2		250	100		350	100
		2	6	5	9		0
	O3			250	150	400	150
		8	3	3	2		0
Demand(D _j):		200	300	350	150	1000	
		0	250	250	0		
			0	0			

		Destination					
		D1	D2	D3	D4	Supply(S _i)	
Source	O1					250	
		3	1	7	4		
	O2					350	
		2	6	5	9		
	O3					400	
		8	3	3	2		
Demand(D _j):		200	300	350	150		

		Variable Cost Table				
		Depots				
Sources		D ₁	D ₂	D ₃	D ₄	D ₅
	O ₁	8	4	3	5	8
	O ₂	3	6	4	8	5
	O ₃	8	4	5	3	4
	O ₄	4	6	8	3	3

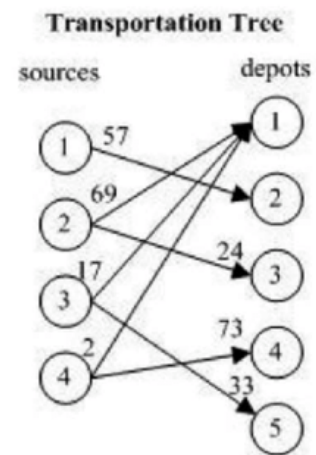
		Fixed Cost Table				
		Depots				
Sources		D ₁	D ₂	D ₃	D ₄	D ₅
	O ₁	60	88	95	76	97
	O ₂	51	72	65	87	76
	O ₃	67	89	99	89	100
	O ₄	86	84	70	92	88

Priority-based encoding chromosome									
node ID	1	2	3	4	1	2	3	4	5
priority	4	1	5	2	3	9	6	8	7

		Transportation Table					
		Depots					
Sources		D ₁	D ₂	D ₃	D ₄	D ₅	a _i
	O ₁		57				57
	O ₂	69		24			93
	O ₃	17				33	50
	O ₄	2			73		75
	b _j	88	57	24	73	33	

a_i: capacity of source i

b_j: demand of depot j



2.2. Potensiallar usulining asosiy g'oyasi

Potensiallar usulida:

- har bir manbaga u_i potensial,
- har bir iste'molchiga v_j potensial
biriktiriladi.

Bazis kataklari uchun:

$$u_i + v_j = c_{ij}$$

Bazis bo'lmagan kataklari uchun baholash qiymati hisoblanadi:

$$\Delta_{ij} = c_{ij} - (u_i + v_j)$$

Agar barcha $\Delta_{ij} \geq 0$ bo'lsa, yechim optimal hisoblanadi

2.3. Potensiallar usulida yechish bosqichlari

1. Boshlang'ich tayanch reja tuziladi

2. Potensiallar (u va v) aniqlanadi
3. Har bir bo'sh katak uchun baholash qiymati hisoblanadi
4. Agar manfiy baholash mavjud bo'lsa, yangi yopiq kontur tuziladi
5. Yuklar qayta taqsimlanadi
6. Optimal holatga erishilguncha jarayon takrorlanadi

2.4. Dasturiy yechimlar bilan solishtirish

Bugungi kunda transport masalasi:

- Python (SciPy, PuLP),
 - MATLAB,
 - Excel Solver
- kabi dasturiy vositalar yordamida ham yechiladi.

Dasturiy yechimlar:

- tezkor;
- katta hajmdagi masalalar uchun qulay;
- avtomatik hisoblash imkonini beradi.

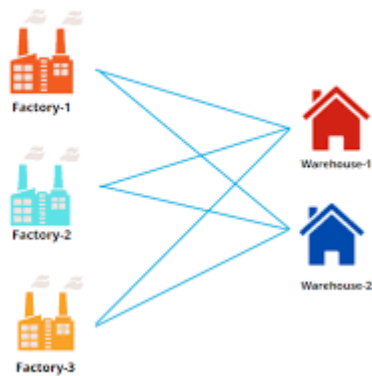
Ammo potensiallar usuli:

- masalaning ichki mantiqini tushunishga yordam beradi;
- nazariy asosni mustahkamlaydi;
- optimal yechimni tekshirish imkonini beradi.

		Destination				Supply(s_i)
		D1	D2	D3	D4	
Source	O1	C_{11}	C_{12}	C_{13}	C_{14}	S_1
	O2	C_{21}	C_{22}	C_{23}	C_{24}	S_2
	O3	C_{31}	C_{32}	C_{33}	C_{34}	S_3
	O4	C_{41}	C_{42}	C_{43}	C_{44}	S_4
Demand(d_j):		d_1	d_2	d_3	d_4	

	A	B	C	D	E	F	G	H	I	J
1	Transportation Problem									
2										
3		Unit Cost	Customer 1	Customer 2	Customer 3					
4		Factory 1	40	47	80					
5		Factory 2	72	36	58					
6		Factory 3	24	61	71					
7										
8										
9		Shipments	Customer 1	Customer 2	Customer 3		Total Out		Supply	
10		Factory 1	0	100	0		100	=	100	
11		Factory 2	0	100	100		200	=	200	
12		Factory 3	200	0	100		300	=	300	
13										
14		Total In	200	200	200					
15			=	=	=				Total Cost	
16		Demand	200	200	200				26000	
17										

Transportation Problem



3. Eng yaxshi yechimni topish va optimal shartlarni tekshirish

3.1. Optimal yechim tushunchasi

Optimal yechim — bu barcha cheklanish shartlarini qanoatlantirgan holda maqsad funksiyasini eng kichik qiymatga olib keluvchi yechimdir.

Transport masalasida optimal yechim:

- barcha talablarni qondiradi;
- barcha zaxiralarni to'liq taqsimlaydi;
- eng kam xarajatni ta'minlaydi.

3.2. Optimal shartni tekshirish

Potensiallar usulida optimal shart quyidagicha:

$$\Delta_{ij} \geq 0 \text{ barcha } i, j \text{ uchun}$$

Agar hech bo'lmaganda bitta $\Delta_{ij} < 0$ bo'lsa, yechim optimal emas va yaxshilanish mumkin.

3.3. Amaliy tekshiruv

Olingan yechim quyidagi mezonlar asosida tekshiriladi:

- barcha $x_{ij} \geq 0$;
- manba va iste'molchi cheklanishlari bajarilgan;
- umumiy xarajat minimal.

Bu tekshiruvlar transport masalasini yechishda muhim bosqich hisoblanadi.

Empty Cells: AF, BD, BF, CD

From \ To	D	E	F	Supply
A	-6 20	+4 30	1	50
B	+3	-8 40	7	40
C	4	4 25	2 35	60
Demand	20	95	35	150

Cell	Closed Loop	Net Cost Change	Opportunity Cost
BD	BD-AD+AE-BE	$3-6+4-8 = -7$	+7
AF	AF-CF+CE-AE	$1-2+4-4 = -1$	+1
BF	BF-CF+CE-BE	$7-2+4-8 = 1$	-1
CD	CD-AD+AE-CE	$4-6+4-4 = -2$	+2

Xulosa

Transport masalasi logistika va texnik tizimlarda muhim ahamiyatga ega bo'lgan optimallashtirish masalasidir. Ushbu mavzuda transport masalasining matematik modeli, potensiallar usuli yordamida yechish va optimal yechimni tekshirish masalalari batafsil yoritildi. Potensiallar usuli nafaqat amaliy hisoblash, balki nazariy tahlil nuqtayi nazaridan ham katta ahamiyatga ega. Ushbu bilimlar talabalarni murakkab optimallashtirish masalalarini tushunish va yechishga tayyorlaydi.

Nazorat savollari

1. Transport masalasi deb nimaga aytiladi?
2. Transport masalasining asosiy elementlarini sanab bering.

3. Transport masalasining maqsad funksiyasi nimani ifodalaydi?
4. Cheklanish shartlari qanday yoziladi?
5. Muvozanatlashgan transport masalasi nima?
6. Potensiallar usuli nima uchun qo'llaniladi?
7. Potensiallar usulida u va v nimani anglatadi?
8. Baholash qiymati qanday hisoblanadi?
9. Optimal yechim qanday shartni qanoatlantirishi kerak?
10. Potensiallar usulini dasturiy yechimlar bilan solishtiring.